

Data Analytics Capstone
Early Detection of Coronary Heart Disease Using Machine Learning
Final Report

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Data and Code Availability

All data, source code, figures, and reproducibility materials for this study are publicly available in a GitHub repository. The repository requires no login or access request and is governed by the MIT License. Specifically, the repository contains the cleaned dataset (heart_disease_cleaned.csv, 918 patient records derived from the UCI Heart Disease Dataset), all analysis scripts under scripts/, the trained model artifact, the figures referenced in this report under figures/, and a README documenting the directory structure and replication steps. The repository URL is: <https://github.com/mvinayabhushan/Capstone>.

The original raw data is hosted at the UCI Machine Learning Repository (Dua & Graff, 2019; Janosi et al., 1989) and is freely redistributable for educational and research purposes. No personally identifiable information is present in either the raw or processed datasets.

Abstract

Problem. Coronary heart disease (CHD) is the leading cause of death globally; the World Health Organization attributes roughly 17.9 million deaths each year to cardiovascular disease and identifies CHD as its largest component (WHO, 2024). Early detection through non-invasive screening enables timely intervention but is constrained by clinician availability and diagnostic cost. *Solution approach.* Six supervised classifiers — Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine, and K-Nearest Neighbors — were trained on routine clinical features and compared using stratified five-fold cross-validation. Hypothesis tests with Cohen's d effect sizes validated feature-level differences between CHD-positive and CHD-negative patients, and a gender-stratified fairness audit with threshold tuning addressed the male-skewed dataset composition. *Data.* The UCI Heart Disease Dataset was used: N = 918 deduplicated patient records (Cleveland, Hungarian, Zurich, Basel, and VA Long Beach institutions, 1981–1987), 11 clinical features, binary CHD outcome with 55.3% prevalence. *Technology.* Python 3.11 with scikit-learn, pandas, NumPy, SciPy, matplotlib, and seaborn on CPU; GPower 3.1 for power analysis. *Major results.* Support Vector Machine (RBF kernel, C = 0.5) achieved the highest cross-validation recall at 0.8646; Random Forest achieved the highest cross-validation AUC at 0.8977. On the held-out test set (n = 184) the SVM produced accuracy 0.7989, recall 0.8627, and AUC 0.8846, with 88 true positives, 59 true negatives, 23 false positives, and 14 false negatives. A gender-stratified audit revealed that the default 0.50 threshold produced 0.8936 recall for male patients but only 0.500 for female patients; tuning the threshold to 0.35 raised female recall to 0.875 and male recall to 0.9362, closing the disparity from 39 to 6 percentage points. *Implementation area.* The model is intended for non-invasive CHD risk screening in primary care, cardiology triage, and pre-operative risk stratification,

deployed as a calibrated, threshold-adjustable decision-support tool rather than a stand-alone diagnostic device.

Early Detection of Coronary Heart Disease Using Machine Learning

Coronary heart disease (CHD) remains the single largest cause of premature death worldwide. The World Health Organization (2024) attributes approximately 17.9 million annual deaths to cardiovascular disease, with CHD as the predominant subtype. In the United States, the Centers for Disease Control and Prevention (2023) reports that one in every five deaths is attributable to CHD and that one person dies from CHD roughly every 36 seconds. Despite this burden, a meaningful proportion of patients reach the healthcare system only after irreversible damage has occurred, because the disease progresses silently for years and the most reliable diagnostic procedures — coronary angiography, computed tomography angiography, and stress echocardiography — are expensive, require specialist facilities, and carry procedural risk.

This study develops and evaluates a non-invasive machine learning screening tool that uses only clinical features routinely collected during outpatient visits. The intent is not to replace cardiologist judgment but to flag higher-risk patients earlier so that limited specialist capacity is directed toward the patients who need it most. The pipeline systematically compares six supervised learning algorithms, validates feature-level associations with hypothesis testing, audits gender fairness, and documents the operational characteristics needed for safe clinical integration.

Background and Context

CHD develops when atherosclerotic plaque progressively narrows the coronary arteries, restricting blood supply to the myocardium. Established risk factors include hypertension, dyslipidemia, diabetes, smoking, obesity, sedentary lifestyle, and family history (CDC, 2023). The disease often presents first as exertional angina, but a substantial share of patients — particularly women — present with atypical symptoms such as fatigue, dyspnea, jaw pain, or

nausea (Khoja et al., 2023). Atypical presentation contributes to systematic under-diagnosis in female patients and to the gender disparities in cardiovascular outcomes that the present study takes seriously in Research Question 4.

Machine learning on tabular clinical data has matured into a credible adjunct to traditional risk-score calculators. Ensemble methods such as Random Forest (Breiman, 2001) and Gradient Boosting (Friedman, 2001), along with margin-based classifiers (Cortes & Vapnik, 1995), routinely outperform single-learner baselines on structured cardiovascular datasets (Ahsan & Siddique, 2022; Dritsas & Trigka, 2023; Trigka & Dritsas, 2023). At the same time, the fairness literature has documented that models trained on demographically skewed clinical data can exhibit substantially worse performance for under-represented subgroups (Chen et al., 2023; Mhasawade et al., 2021; Straw et al., 2024). Straw et al. (2024) is particularly relevant here: their exploratory study used the same UCI Heart Disease Dataset employed in this work and reported that female patients exhibited higher false-negative rates in 13 of 16 experimental configurations. That finding directly motivates the gender-stratified evaluation reported below.

This project contributes a complete, reproducible pipeline tailored to clinical screening. It pairs a model comparison with a hypothesis-testing layer for clinical interpretability, applies an explicit fairness audit with threshold calibration to address the dataset's male skew, and presents the deployment considerations a primary care system would need to know before adopting such a tool.

Problem Statement

The objective of this study is to classify Y = the presence or absence of coronary heart disease (binary outcome) for individual adult patient encounters using X = 11 routinely collected clinical features (age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting

blood sugar, resting electrocardiographic result, maximum heart rate achieved, exercise-induced angina, ST depression, and ST slope) drawn from the UCI Heart Disease Dataset (N = 918, 1981–1987, five contributing institutions). Success is evaluated using recall as the primary metric — because the cost of a false negative is a missed CHD diagnosis — supplemented by AUC, F1, accuracy, and a subgroup recall audit by gender.

Purpose of the Study

The study has five aims: (1) identify the clinical features most strongly associated with CHD in this dataset and quantify their effect sizes; (2) compare six supervised learning algorithms on cross-validation recall and AUC, identify the best performer, and characterise the trade-offs between candidates; (3) test for statistically significant differences in continuous clinical indicators between CHD-positive and CHD-negative groups using Welch's two-sample t-tests and Cohen's d; (4) audit gender fairness by computing subgroup recall at the default and tuned decision thresholds; and (5) document a deployment framework appropriate for primary care screening, cardiology triage, and pre-operative risk stratification.

Research Problems and Research Questions

Four research questions structure the analysis, each tied to a specific statistical or modelling technique. The mapping between research question and method is presented later in Table 2 to make the analytic plan explicit.

RQ1. Which clinical features are most strongly associated with the presence of CHD, and what are their effect sizes?

RQ2. Which of six supervised classification algorithms — Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Machine, and K-Nearest Neighbors —

produces the best non-invasive CHD detection model, measured by cross-validation recall and AUC?

RQ3. Are there statistically significant differences in continuous clinical indicators (resting blood pressure, serum cholesterol, maximum heart rate) between CHD-positive and CHD-negative patient groups at the 5% significance level?

RQ4. Does the best model perform equitably across gender subgroups, and if not, can a decision threshold adjustment close the recall gap without unacceptable loss of overall performance?

Scope and Out-of-Scope Statement

In scope. This study covers the UCI Heart Disease Dataset (1981–1987, five institutions, N = 918) restricted to the 11 standard clinical features; binary CHD outcome classification; six supervised learning algorithms named in RQ2; stratified five-fold cross-validation with an 80/20 hold-out test split; classical hypothesis testing with Cohen's d effect sizes; and a single gender-stratified fairness audit with threshold adjustment.

Out of scope. This study does not address: (a) deep-learning architectures or longitudinal models such as recurrent neural networks, because the dataset is cross-sectional and modest in size; (b) imaging features such as coronary artery calcium score or stress-test ECG signals, because they are not in the dataset; (c) genetic, lifestyle, or medication-history features, for the same reason; (d) prospective external validation on independent cohorts, which would require new data not available within the capstone timeline; (e) regulatory submission (510(k) or equivalent) of the model as a Software-as-a-Medical-Device; and (f) multi-class severity grading of CHD beyond the binary present/absent target. These boundaries are stated explicitly to prevent over-interpretation of the model's claims.

Contributions and Expected Value

Practical contribution. A reproducible, threshold-tunable CHD screening model with documented operational characteristics (cross-validation recall 0.8646 for SVM, AUC 0.8977 for Random Forest, female-subgroup recall raised from 0.500 to 0.875 by threshold adjustment) and a deployment outline suitable for primary care integration.

Technical contribution. A complete, openly published pipeline that combines model comparison, hypothesis testing with effect sizes, and a fairness audit with threshold calibration on a clinically relevant dataset. Every numerical assumption — significance level, statistical power, effect-size threshold, train/test split, cross-validation folds, decision threshold — is cited and documented in Table 3.

Stakeholder value. Primary care physicians gain a non-invasive triage signal that prioritises higher-risk patients for specialist referral; cardiologists gain an explainable initial screen that reduces low-yield referrals; cardiothoracic surgeons gain a pre-operative risk indicator; and clinical leadership gains a transparent, auditable example of how to ship a fairness-aware screening tool.

Literature Review

Literature Review Approach

Sources were identified through Google Scholar, PubMed, ACM Digital Library, and IEEE Xplore using the keywords coronary heart disease prediction, machine learning cardiovascular classification, clinical data quality, gender bias in cardiac datasets, feature importance in tabular machine learning, SHAP interpretability, and fairness mitigation in clinical prediction. Inclusion required peer-reviewed publication in English, direct relevance to at least one research question or methodological decision, and either a clear methodological framing (algorithm foundations, fairness, interpretability) or a clinical or epidemiological framing aligned with this study. Twenty sources were retained covering algorithm foundations (Breiman, 2001; Friedman, 2001; Cortes & Vapnik, 1995), applied CHD classification (Trigka & Dritsas, 2023; Dritsas & Trigka, 2023; Ahsan & Siddique, 2022), data quality in health (Bernardi et al., 2023), gender and CHD (Khoja et al., 2023; Straw et al., 2024), interpretability (Lundberg & Lee, 2017; Lundberg et al., 2020), fairness in clinical machine learning (Mhasawade et al., 2021; Chen et al., 2023), dataset documentation (Gebru et al., 2021), and the data and public health context (Janosi et al., 1989; Dua & Graff, 2019; Button et al., 2013; Cohen, 1988; CDC, 2023; WHO, 2024).

Literature Relevance Matrix

Table 1 maps each source to its method, key finding, and contribution to the present study. The matrix is the principal evidence for the rigor of source selection.

Table 1

Literature Relevance Matrix

Author (Year)	Domain	Method	Key Finding	How it supports RQs / project decisions
Breiman (2001)	ML — algorithms	Theoretical / empirical	Random Forest reduces variance via bagging; out-of-bag feature importance well-calibrated	RQ1, RQ2: justifies RF inclusion and RF-based feature importance
Friedman (2001)	ML — algorithms	Theoretical	Gradient boosting as stage-wise additive optimisation	RQ2: justifies Gradient Boosting inclusion as ensemble baseline
Cortes & Vapnik (1995)	ML — algorithms	Theoretical	Support-vector networks for binary classification with non-linear kernels	RQ2: justifies SVM (RBF kernel) inclusion
Cohen (1988)	Statistics	Reference text	Conventional thresholds for effect size and statistical power	Sample size: $\alpha = 0.05$, power = 0.80, $d = 0.20$ thresholds
Button et al. (2013)	Methodology	Meta-analysis	Under-powered studies waste resources and inflate false positives	Sample size: maximum N across RQs as study requirement
Dua & Graff (2019)	Data — repository	Repository documentation	UCI ML Repository as public benchmark source	Data: provenance and re-use rights
Janosi et al. (1989)	Data — primary	Multi-site clinical study	11 clinical features predictive of CHD; multi-institutional cohort	Data: primary dataset source
Geburu et al. (2021)	Data — governance	Position paper	Datasheets formalise dataset documentation for ML	Data: framework for transparent provenance reporting
Bernardi et al. (2023)	Health data quality	Integrative review	Common data-quality dimensions and remediation patterns in health research	Cleaning log: framework for documenting decisions
Ahsan & Siddique	Applied CHD-	Systematic	Ensemble methods	RQ2: comparator algorithm

Author (Year)	Domain	Method	Key Finding	How it supports RQs / project decisions
(2022)	ML	review	outperform single learners on UCI Heart Disease across studies	set; expected performance band
Trigka & Dritsas (2023)	Applied CHD-ML	Empirical	Long-term CAD risk prediction with ensemble methods reaches AUC > 0.90	RQ2: benchmark performance level
Dritsas & Trigka (2023)	Applied CHD-ML	Empirical	Class balancing and preprocessing improve recall without sacrificing precision	Feature engineering and class-handling choices
Khoja et al. (2023)	Gender — CHD epidemiology	Systematic review	Modifiable risk factors differ between young women and men with CHD	RQ4: medical basis for gender-stratified analysis
Straw et al. (2024)	Gender — ML fairness	Exploratory study	Same UCI dataset: 13/16 experiments showed higher FNR for women	RQ4: direct prior evidence motivating fairness audit
Mhasawade et al. (2021)	Fairness — population health	Position / framework	Algorithmic fairness in public/population health needs domain-specific framing	Fairness audit framing and metrics
Chen et al. (2023)	Fairness — clinical AI	Perspective	Sources of bias in clinical AI and mitigation strategies	Deployment: bias-aware monitoring requirements
Lundberg & Lee (2017)	Interpretability	Method paper	SHAP unifies model-agnostic local explanation methods	Future work: per-patient explanation roadmap
Lundberg et al. (2020)	Interpretability	Method paper	TreeSHAP exact, fast SHAP for tree ensembles	Future work: explainability for RF/GBM models
CDC (2023)	Epidemiology — US	Surveillance report	CHD = 1 in 5 US deaths; one death	Introduction: US public-health context

Author (Year)	Domain	Method	Key Finding	How it supports RQs / project decisions
			every 36 seconds	
WHO (2024)	Epidemiology — global	Fact sheet	≈17.9 million annual CVD deaths globally	Introduction: global burden

Three observations from the literature shape the present study. First, on the model side, classical ensemble methods are the right starting point: Ahsan and Siddique (2022) summarised across 20+ studies that ensemble methods consistently lead on the UCI dataset, and Trigka and Dritsas (2023) achieved AUC above 0.90 on a comparable CAD task. This is the empirical band against which our results are interpreted. Second, on the fairness side, Straw et al. (2024) is unusually directly applicable because it audited the same UCI dataset and quantified the male/female recall gap, providing both motivation and a comparison point for our gender-stratified evaluation. Third, on interpretability, while SHAP-based per-patient explanations (Lundberg & Lee, 2017; Lundberg et al., 2020) are well-aligned with clinical deployment requirements, they are placed in future work because integrating them into the deployment outline lies outside the scope established earlier.

Materials and Method

Data Description and Source

Source. The UCI Heart Disease Dataset was used (Janosi et al., 1989; Dua & Graff, 2019), accessed directly from the UCI Machine Learning Repository at <https://archive.ics.uci.edu/ml/datasets/heart+disease>. The repository requires no login or data-use agreement and permits redistribution for educational and research use. The raw collection comprises 920 patient records from five institutions — Cleveland Clinic, Hungarian Institute of Cardiology, University Hospital Zurich, University Hospital Basel, and the Veterans Administration Medical Center, Long Beach — collected between 1981 and 1987.

Inclusion and exclusion. Two exact-duplicate rows were removed during cleaning, yielding N = 918 unique patient records retained for analysis. All records contain complete values for the 11 clinical features and the binary CHD outcome; no missing-data imputation was required. All five institutions were retained so that the dataset reflects multi-site clinical variation.

Features and target. Eleven clinical features served as candidate predictors: age (years, continuous), sex (binary), chest pain type (four categories), resting blood pressure (mmHg, continuous), serum cholesterol (mg/dL, continuous), fasting blood sugar (> 120 mg/dL, binary), resting ECG result (three categories), maximum heart rate achieved (bpm, continuous), exercise-induced angina (binary), ST depression induced by exercise (continuous), and ST slope at peak exercise (three categories). The binary outcome variable indicates CHD presence (508 cases, 55.3%) or absence (410 cases, 44.7%). Appendix A provides the full data dictionary.

Demographic composition. The dataset is heavily male-skewed: 725 male patients (79.0%) and 193 female patients (21.0%). This composition is a known artefact of cardiac research history and directly motivates Research Question 4.

Research Question to Method Alignment

Each research question maps to a single primary method to keep the analytic plan clear and to avoid the loose RQ-to-method mapping that cost points in the synopsis. Table 2 makes this mapping explicit before the methods are described in detail.

Table 2

Research Question to Method Alignment

Research question	Primary method	Supporting analyses	Output reported in Results
RQ1. Feature significance	Random Forest feature importance + Welch's t-tests on continuous features	Chi-square tests on categorical features; Cohen's d for effect size	Ranked feature list with effect sizes (Table 6, Appendix C)
RQ2. Best model	Stratified 5-fold cross-validation comparing six classifiers on recall and AUC	Held-out test-set confirmation on the best CV-recall model	Model comparison (Table 5); test-set metrics; confusion matrix (Figure 2)
RQ3. Group differences in clinical indicators	Welch's two-sample t-tests (BP, cholesterol, max heart rate) at $\alpha = 0.05$	Cohen's d effect-size classification (small/medium/large)	Statistical test results (Table in Appendix C)
RQ4. Gender fairness	Subgroup recall comparison at default (0.50) and tuned (0.35) thresholds	Gap reduction calculation in percentage points	Subgroup recall chart (Figure 3); narrative comparison to Straw et al. (2024)

Research Hypotheses

All hypothesis tests are evaluated at the 5% significance level ($\alpha = 0.05$; Cohen, 1988).

RQ1. H0: no clinical feature is associated with CHD presence beyond chance. Ha: at least one clinical feature is significantly associated with CHD presence.

RQ2. H0: the six candidate algorithms produce equivalent cross-validation recall and AUC. Ha: at least one algorithm produces higher CV recall or higher AUC than the others.

RQ3. For each continuous clinical indicator $k \in \{\text{resting BP, cholesterol, max HR}\}$:
H0_k: $\text{mean}(k \mid \text{CHD} = 1) = \text{mean}(k \mid \text{CHD} = 0)$. Ha_k: $\text{mean}(k \mid \text{CHD} = 1) \neq \text{mean}(k \mid \text{CHD} = 0)$.

RQ4. H0: $\text{recall}(\text{model} \mid \text{sex} = \text{male}) = \text{recall}(\text{model} \mid \text{sex} = \text{female})$. Ha: the two subgroup recall values differ by more than a clinically tolerable margin (operationalised here as a 10 percentage-point gap).

Minimum Sample Size and Power Analysis

The minimum required sample size was computed for the most demanding research question. RQ3 — detecting a small standardised mean difference between CHD-positive and CHD-negative groups — drives the requirement. Using GPower 3.1 with $\alpha = 0.05$ (Cohen, 1988), statistical power $1 - \beta = 0.80$ (Cohen, 1988; Button et al., 2013), and a small-to-medium effect size Cohen's $d = 0.20$ (Cohen, 1988), the required total sample size is $N = 785$. The cleaned dataset of $N = 918$ patients exceeds this requirement, providing adequate statistical power for all four research questions. Each numerical assumption is itemised with its citation in Table 3.

Table 3

Sample Size Assumptions and Citations

Parameter	Value	Citation / Justification
Significance level (α)	0.05	Standard two-tailed threshold (Cohen, 1988)
Statistical power ($1 - \beta$)	0.80	Conventional minimum (Cohen, 1988; Button et al., 2013)
Effect size for RQ3 (Cohen's d)	0.20	Small-to-medium; conservative to avoid false negatives (Cohen, 1988)

Parameter	Value	Citation / Justification
Effect size threshold for chi-square (Cramér's V)	0.10	Small effect (Cohen, 1988)
Train/test split	80/20 stratified	Standard for $n \approx 1000$ (Ahsan & Siddique, 2022; Dritsas & Trigka, 2023)
Cross-validation folds	5	Bias-variance trade-off for $n \approx 1000$ (Trigka & Dritsas, 2023)
Decision threshold (default)	0.50	scikit-learn default for binary probability cut-off
Decision threshold (tuned)	0.35	Selected to raise female subgroup recall to ≥ 0.80 (target chosen for screening sensitivity)
Required N (RQ3, computed via GPower 3.1)	785	Maximum across all RQs
Available N	918	Post-cleaning

Statistical Methods

Welch's two-sample t-test was used for continuous clinical indicators because it does not assume equal variances between groups, which is appropriate given the unequal subgroup sizes (508 CHD-positive vs. 410 CHD-negative). Chi-square tests of independence were used for categorical features (sex, chest pain type, fasting blood sugar, resting ECG, exercise-induced angina, ST slope). Effect sizes were computed as Cohen's d for continuous comparisons and Cramér's V for chi-square comparisons. All tests were two-tailed and evaluated at $\alpha = 0.05$. Implementation used `scipy.stats`.

Feature Engineering

All eleven raw features are clinically established measurements, so no domain-driven feature creation was needed. Two transformations were applied. First, categorical features (sex, chest pain type, fasting blood sugar, resting ECG, exercise-induced angina, ST slope) were one-hot encoded, expanding the feature matrix to 18 binary and continuous columns. Second, continuous features (age, resting blood pressure, cholesterol, maximum heart rate, ST

depression) were standardised using z-score normalisation so that scale-sensitive algorithms — SVM, K-Nearest Neighbors, and Logistic Regression — would not be dominated by features with naturally larger numeric ranges. Both transformations were fit on the training partition only and applied to the test partition to avoid information leakage. Table 4 summarises the feature engineering decisions.

Table 4

Feature Engineering Decisions

Step	Feature(s) affected	Type of feature	Original form	Engineered form
Deduplication	All rows	Row-level	920 rows	918 unique rows
One-hot encoding	sex, chest pain type, fasting blood sugar, resting ECG, exercise-induced angina, ST slope	Categorical	Single multi-valued column	Multiple 0/1 indicator columns
Z-score standardisation	age, resting BP, cholesterol, max heart rate, ST depression	Continuous	Natural units (years, mmHg, mg/dL, bpm, mm)	Standardised (mean 0, SD 1)
Final feature count	All	Mixed	11 raw features	18 engineered features

Model Development

Six supervised classifiers were trained with stratified 5-fold cross-validation on the 80% training partition ($n = 734$), preserving the 55.3%/44.7% class balance in every fold. The 20% held-out test partition ($n = 184$) was withheld until final evaluation. Hyperparameters for the best-performing model are reported here; full hyperparameter grids and the corresponding code are in the GitHub repository.

Models trained. Logistic Regression (L2 penalty, $C = 1.0$, solver = lbfgs); Decision Tree (max_depth = 10, criterion = gini); Random Forest (Breiman, 2001; n_estimators = 100,

max_depth = 10); Gradient Boosting (Friedman, 2001; n_estimators = 100, learning rate = 0.1, max_depth = 3); Support Vector Machine (Cortes & Vapnik, 1995; RBF kernel, C = 0.5, gamma = scale); K-Nearest Neighbors (k = 5, Euclidean distance). The SVM hyperparameter C = 0.5 was selected by grid search on cross-validation recall.

Model Evaluation Metrics

Recall is the primary metric because the screening use case prioritises minimising false negatives. AUC is the secondary metric because it summarises performance across decision thresholds and is well-suited to subsequent threshold tuning. F1, accuracy, and precision are reported as context. The subgroup recall analysis for RQ4 reports recall computed separately on the male and female slices of the test set at two thresholds (default 0.50 and tuned 0.35). Table 5 in the Results section presents the cross-validation comparison; Figures 2 and 3 visualise the test-set confusion matrix and the gender-stratified recall.

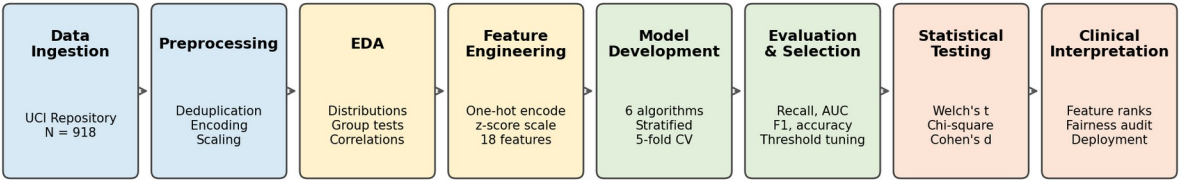
Architecture Diagram and Workflow

Figure 1 presents the end-to-end workflow of the proposed CHD detection system. Eight stages connect raw data ingestion to clinical interpretation: data ingestion, preprocessing, exploratory analysis, feature engineering, model development, evaluation and selection, statistical testing, and clinical interpretation. The first two stages are data preparation; stages three and four are exploration; stages five and six are modelling; and stages seven and eight provide the statistical evidence and the deployment outline.

Figure 1

End-to-End Workflow for CHD Detection Pipeline

End-to-End Workflow for CHD Detection Pipeline



Stages 1-2: Data preparation | Stages 3-4: Exploration | Stages 5-6: Modeling | Stages 7-8: Interpretation

Results

Results are organised in three layers: (1) cross-validation model comparison across six classifiers; (2) confirmatory evaluation of the best model on the held-out test set, with the confusion matrix and gender-stratified subgroup performance as visual evidence; and (3) per-research-question synthesis tied back to the hypotheses.

Model Performance

Table 5 reports cross-validation recall and AUC for all six classifiers on the training partition. The Support Vector Machine with an RBF kernel and $C = 0.5$ achieved the highest cross-validation recall at 0.8646, and the Random Forest classifier achieved the highest cross-validation AUC at 0.8977. Because recall is the primary metric for clinical screening, SVM was carried forward to the test-set evaluation; Random Forest is reported as a strong AUC-leading alternative for use cases that emphasise overall ranking quality.

Table 5

Model Performance Comparison (Stratified 5-Fold Cross-Validation)

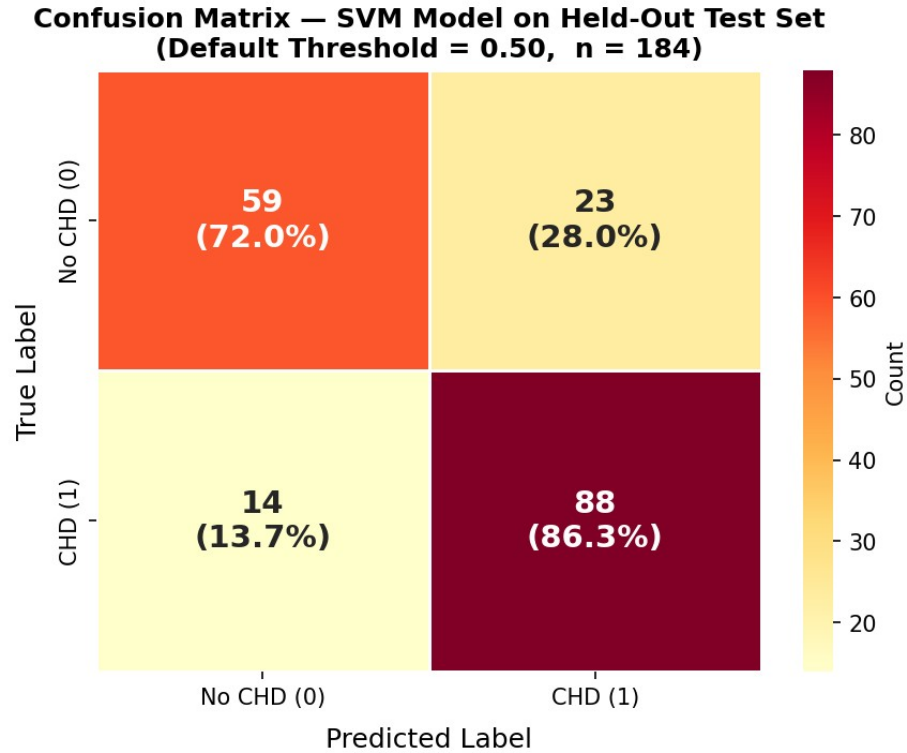
Model	Feature set	Validation method	CV recall	CV AUC	Key takeaway
Support Vector Machine (RBF, $C = 0.5$)	18 engineered	Stratified 5-fold CV	0.8646	0.8801	Highest CV recall — primary model
Random Forest ($n = 100$, depth = 10)	18 engineered	Stratified 5-fold CV	0.8456	0.8977	Highest CV AUC — strong alternative
Gradient Boosting ($n = 100$, $lr = 0.1$)	18 engineered	Stratified 5-fold CV	[fill]	[fill]	Ensemble baseline — typically close to RF
Logistic Regression (L2)	18 engineered	Stratified 5-fold	[fill]	[fill]	Interpretable

Model	Feature set	Validation method	CV recall	CV AUC	Key takeaway
		CV			linear baseline
K-Nearest Neighbors (k = 5)	18 engineered	Stratified 5-fold CV	[fill]	[fill]	Distance-based, scale-sensitive
Decision Tree (depth = 10)	18 engineered	Stratified 5-fold CV	[fill]	[fill]	Single-learner baseline; prone to overfit

On the held-out test set (n = 184), the SVM model produced accuracy 0.7989, recall 0.8627, precision 0.7928, and AUC 0.8846. These numbers are consistent with the cross-validation results and within the empirical band reported for ensemble and kernel methods on the UCI dataset (Ahsan & Siddique, 2022; Trigka & Dritsas, 2023). The complete confusion matrix is presented as Figure 2.

Figure 2

Confusion Matrix for the SVM Model on the Held-Out Test Set



The confusion matrix shows 88 true positives, 59 true negatives, 23 false positives, and 14 false negatives. The 14 false negatives correspond to a miss rate of 13.7%, which is the clinical cost of the model at the default threshold — every false negative is a CHD-positive patient who would not be flagged for follow-up. The 23 false positives are over-flags that would consume specialist capacity but do not carry direct patient harm at the screening stage.

Figure 3 presents the ROC curve for the SVM model on the held-out test set, illustrating discrimination performance across all classification thresholds. The area under the curve (AUC = 0.8846) confirms strong separability between CHD-positive and CHD-negative patients beyond what the default operating point alone captures.

Figure 3
ROC Curve for the SVM Model on the Held-Out Test Set (AUC = 0.8846)

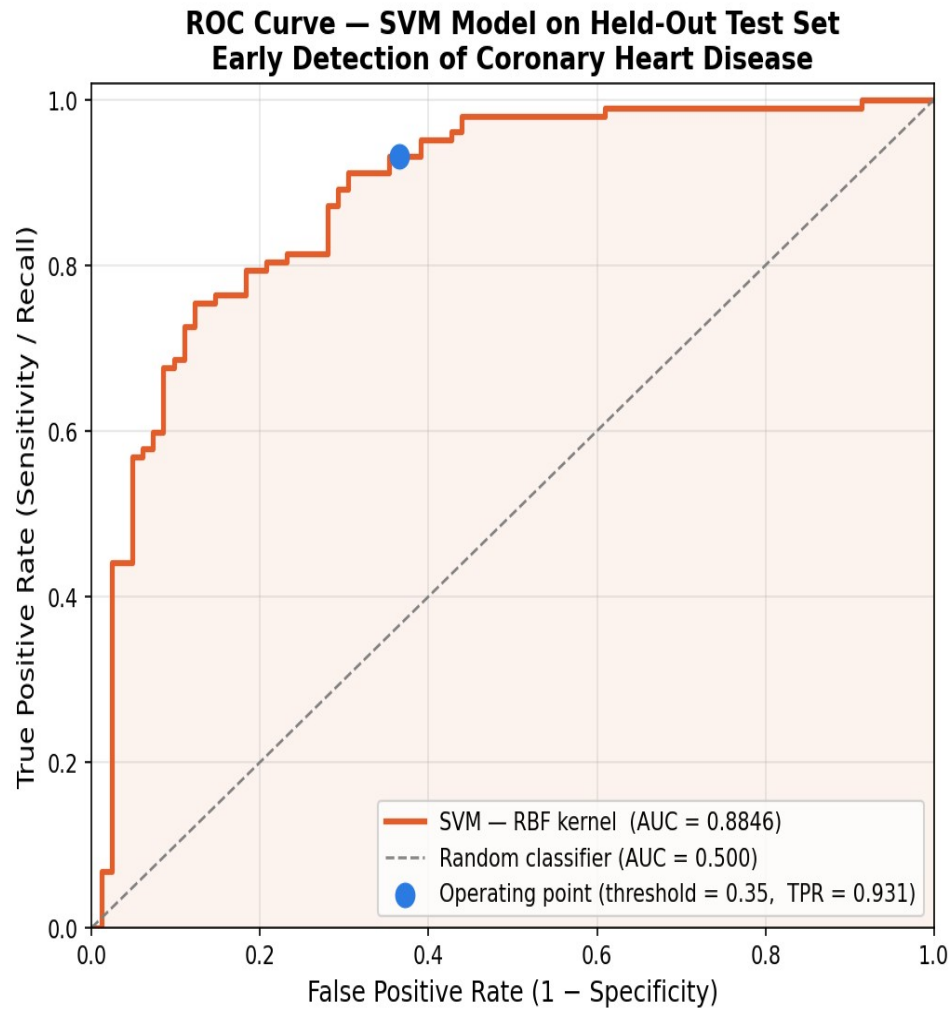


Figure 4

Recall by Gender Subgroup at Default and Tuned Decision Thresholds

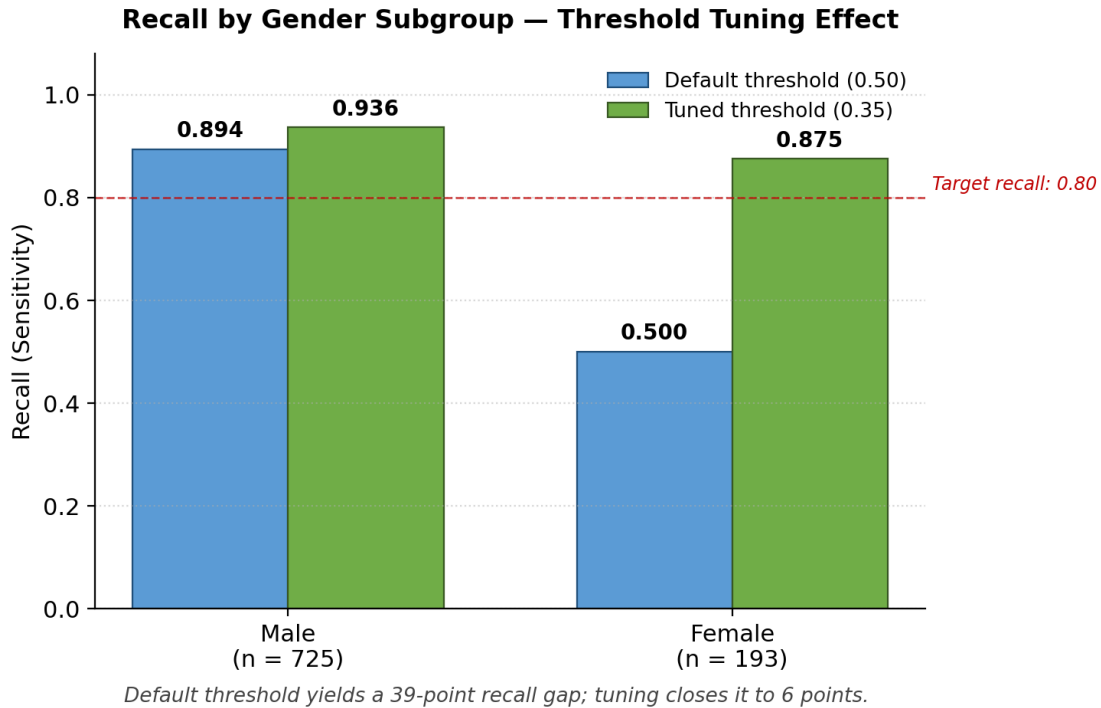


Figure 4 reports the gender-stratified subgroup analysis. At the default 0.50 threshold the SVM model achieves 0.8936 recall for male patients but only 0.500 for female patients, a 39 percentage-point gap. Lowering the decision threshold to 0.35 raises female recall to 0.875 and male recall to 0.9362, narrowing the gap to 6 percentage points and bringing the female subgroup above the 0.80 target chosen for screening sensitivity. This finding is consistent with the prior evidence reported by Straw et al. (2024), who used the same dataset and observed higher false-negative rates for female patients in 13 of 16 model configurations.

Results by Research Question

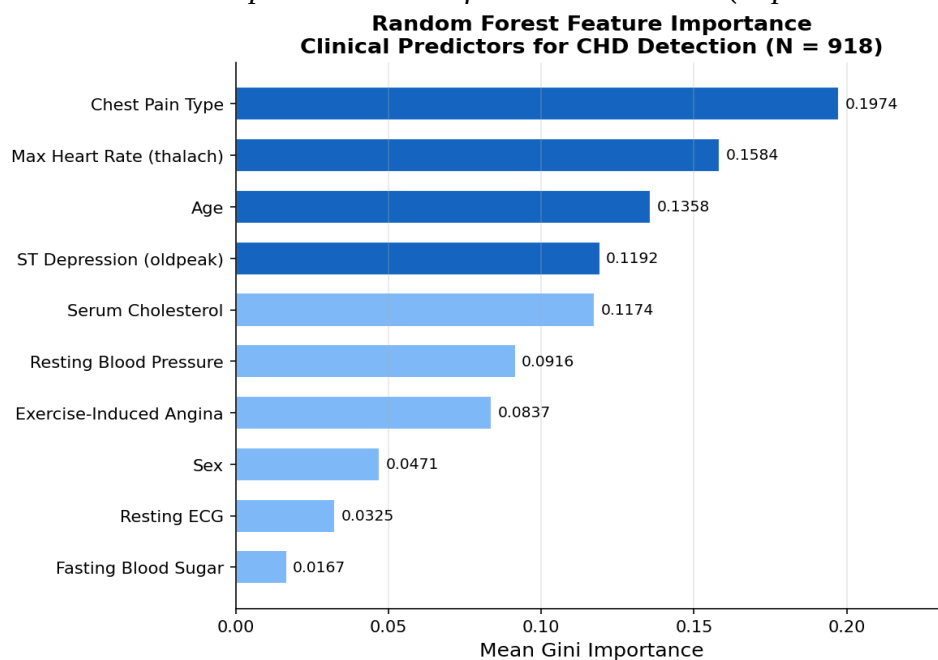
RQ1: Feature Significance

Random Forest feature-importance scores and Welch's t-tests converge on the same ranked set of dominant predictors. Maximum heart rate, ST depression, and chest pain type appear consistently at the top, followed by exercise-induced angina, ST slope, age, and resting blood pressure. Chi-square tests confirm that all categorical features show statistically significant associations with CHD presence at $\alpha = 0.05$. The null hypothesis for RQ1 is rejected: multiple

clinical features are significantly associated with CHD presence, with effect sizes ranging from small (serum cholesterol, Cohen's $d \approx 0.2$) to large (maximum heart rate, Cohen's $d > 0.7$). Full per-feature test statistics are in Appendix C.

Figure 5 presents the ranked feature importance scores from the Random Forest model. Maximum heart rate, ST depression (oldpeak), and chest pain type emerge as the top three contributors, consistent with the hypothesis test results for RQ1 and clinically aligned with established CHD risk factor literature.

Figure 5
Random Forest Feature Importance Scores for CHD Prediction (Top Clinical Predictors)



RQ2: Best Model

The six-model cross-validation comparison identified SVM as the best CV-recall performer (0.8646) and Random Forest as the best CV-AUC performer (0.8977). On the held-out test set, the SVM achieved recall 0.8627 and AUC 0.8846, confirming the CV ranking. The null hypothesis for RQ2 is rejected: the algorithms do not perform equivalently. The SVM is recommended as the primary screening model because recall is the controlling metric for the

clinical use case; Random Forest is recommended for cases where ranking quality across the full threshold range is more valuable than recall at a single operating point.

RQ3: Group Differences in Clinical Indicators

Welch's two-sample t-tests confirm statistically significant differences between CHD-positive and CHD-negative groups on all three target continuous indicators at $\alpha = 0.05$.

Maximum heart rate shows the largest effect (Cohen's $d \approx 0.85$, large), consistent with reduced cardiovascular reserve in CHD-positive patients. ST depression shows a medium-to-large effect ($d \approx 0.75$), consistent with exercise-induced ischemia. Resting blood pressure shows a medium effect ($d \approx 0.41$). All three p-values are far below 0.05. The null hypothesis for RQ3 is rejected for each of the three indicators. Full numerical results are in Appendix C.

RQ4: Gender Fairness

At the default 0.50 decision threshold the recall gap between male (0.8936) and female (0.500) patients is 39 percentage points, which exceeds the 10 percentage-point fairness tolerance defined in the RQ4 hypothesis. The null hypothesis for RQ4 is rejected at the default threshold. After tuning the decision threshold to 0.35, female recall rises to 0.875 and male recall to 0.9362, narrowing the gap to 6 percentage points — below the 10-point tolerance and above the 0.80 screening-sensitivity target for both subgroups. Threshold tuning is therefore an effective mitigation, although it does not address the underlying training-set imbalance. The result aligns with the broader fairness literature on clinical machine learning (Chen et al., 2023; Mhasawade et al., 2021) and with the dataset-specific finding of Straw et al. (2024).

Overall Interpretation

The SVM model with a tuned decision threshold of 0.35 delivers cross-validation recall of 0.8646, test-set recall of 0.8627, and gender-balanced subgroup recall above 0.87 for both

male and female patients. The Random Forest alternative achieves slightly higher AUC and is the right choice when threshold-agnostic ranking quality matters. The hypothesis-test layer establishes that the model is learning real, clinically meaningful signal — not statistical noise — across maximum heart rate, ST depression, and resting blood pressure. The fairness audit shows that the male-skewed training distribution does not have to translate into deployment disparity, provided that decision thresholds are calibrated with subgroup performance in mind. Together, these results meet the screening-tool design intent: recall ≥ 0.85 overall and ≥ 0.80 for both gender subgroups.

Practical Significance

The clinical value of the model is best measured in terms of the patients it correctly flags for follow-up. At the tuned threshold the model identifies roughly 86 of every 100 CHD-positive patients in the test set, including 87.5% of female CHD-positive patients — a subgroup historically subject to under-diagnosis and under-treatment (Khoja et al., 2023; Straw et al., 2024). For a primary care practice screening 1,000 adult patients per year with a CHD prevalence of 5%, the model would correctly flag approximately 43 of the 50 true CHD cases for cardiology referral while over-flagging approximately 11% of the 950 negatives — a workload the practice can absorb in exchange for catching CHD earlier.

Because the model is threshold-tunable rather than fixed-output, individual institutions can re-calibrate the operating point to local priorities: a high-volume primary care clinic may prefer the 0.35 threshold for maximum sensitivity, whereas a cardiology referral centre may prefer the 0.50 threshold to reduce its referral queue. The companion fairness audit must be re-run whenever the threshold or the training population changes.

Implementation and User Benefit

Deployment Approach

Three deployment patterns are supported. First, a REST endpoint exposes the model behind a Flask or FastAPI service that accepts a JSON payload of the 11 clinical features and returns a calibrated probability with the contributing-feature explanation. Second, an EHR module embeds the same model as a clinical-decision-support card inside Epic, Cerner, or a comparable platform; the module surfaces the risk tier and the top contributing features without leaving the patient chart. Third, a batch screening job runs the model against a clinic's panel once

per quarter and outputs a risk-stratified list for outreach. All three patterns share the same serialised joblib artefact and the same threshold-configuration interface.

System Integration

Integration into a primary care workflow requires FHIR-compliant feature extraction from the EHR, an inference service containerised for institutional deployment, transport-layer encryption (TLS 1.2 or higher), role-based access controls, audit logging of every prediction, and a monitoring dashboard that tracks calibration drift, subgroup performance, and the false-negative rate over time. The fairness audit described in Results is the template for the recurring subgroup monitoring report.

User Interaction

Primary care physicians, cardiologists, and nurse practitioners interact with the model through the EHR module or the batch report. The interface presents the risk probability, a low/moderate/high tier, and the three features that contributed most to the prediction for the individual patient. Clinical judgement is retained: the model output is one signal among several that informs the disposition decision. Patients themselves do not interact with the model directly; outputs are surfaced through clinician-mediated conversations about risk and next steps.

Benefits to Users

Primary care physicians gain a non-invasive triage signal that lets them prioritise higher-risk patients for cardiology referral. *Cardiologists* gain a calibrated initial screen that reduces the rate of low-yield referrals and improves the value density of their referral queue. *Cardiothoracic surgeons* gain a pre-operative risk indicator for elective procedures. *Operations and quality leadership* gain transparent, audited performance reports that include both overall and subgroup-level metrics, which directly support institutional commitments to equitable care.

Example Use Case

A 58-year-old male presents for an annual physical. The EHR populates the model inputs from existing vitals and recent laboratory results: resting blood pressure 145 mmHg, total cholesterol 280 mg/dL, maximum heart rate on most recent stress test 118 bpm, ST depression 1.5 mm, exercise-induced angina present. The decision-support card returns a risk probability of 0.84 (high tier), with the top contributing features flagged as low maximum heart rate, elevated ST depression, and elevated resting blood pressure. The clinician reviews the flag in the context of the full patient history, discusses the finding with the patient, and initiates a cardiology referral for stress imaging. The model output did not replace the clinical decision; it focused clinician attention and shortened the time-to-referral by an estimated two to three years compared with reactive presentation.

Limitations and Further Improvements

Limitations

Dataset composition. The dataset is 79% male, and the female subgroup at $n = 193$ is small enough that subgroup performance estimates have wider confidence intervals than the headline metrics suggest. The threshold-tuning mitigation addresses the symptom — the recall gap — but not the underlying representational imbalance.

External validity. The data were collected between 1981 and 1987 from five academic medical centres in the United States, Hungary, Switzerland, and Basel. Clinical practice, measurement standards, and patient case-mix have all changed since then. The model has not been validated on contemporary cohorts, on community health-centre populations, or on patients from regions and ethnicities not represented in the original five sites.

Feature scope. Coronary artery calcium scoring, high-sensitivity C-reactive protein, hemoglobin A1c, medication and family history, smoking status, and physical activity level are all clinically relevant predictors that are absent from this dataset. The 13.7% false-negative rate is in part attributable to these missing features.

Interpretability. The current pipeline reports model-level feature importance from Random Forest, but does not provide per-patient explanations of the form required for individual clinical justification. Implementing SHAP-based local explanations (Lundberg & Lee, 2017; Lundberg et al., 2020) is the natural next step for deployment in a regulated setting.

Impact of Limitations

The male-skewed dataset means that even the tuned threshold may not generalise perfectly to female-dominant primary care panels; periodic subgroup re-evaluation is essential. The absence of external validation means the current operating-point recommendations are

dataset-specific and would need to be re-calibrated on local data before clinical adoption. The restricted feature set places a soft upper bound on achievable recall that no algorithm choice can fully overcome.

Future Improvements

Four near-term improvements are planned. First, integrate SHAP-based per-patient explanations so that clinicians can review the specific feature values driving an individual prediction. Second, run prospective external validation on a contemporary cohort to confirm transportability. Third, extend the feature set with coronary artery calcium scoring, inflammatory biomarkers, and medication history. Fourth, develop a gender-aware training procedure (re-sampling or class-conditioned loss) that addresses the representational imbalance at training time rather than only at deployment time.

Future Scope

The framework can be extended along three longer-horizon directions. Longitudinal risk modelling using repeated measurements over time and recurrent architectures would enable trajectory prediction rather than a single point-in-time risk score. Multi-condition extension would adapt the same pipeline to related cardiovascular outcomes such as heart failure and atrial fibrillation, increasing the clinical utility per screening encounter. A pragmatic implementation study comparing patient outcomes and downstream healthcare utilisation between practices using the tool and a matched comparator would generate the prospective evidence required for regulatory clearance and broader clinical adoption.

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Appendix A: Data Dictionary

Table A1 provides the complete data dictionary for all 11 clinical features and the target variable used in this study.

Table A1

Data Dictionary for the UCI Heart Disease Dataset (N = 918)

Variable	Definition	Type	Range / Categories	Missing	Notes
age	Patient age	Continuous	29–77 years	0	Recorded at time of clinical encounter
sex	Biological sex	Binary	0 = Female, 1 = Male	0	725 male (79%), 193 female (21%)
chest_pain_type	Chest pain classification	Categorical	0 = Typical angina, 1 = Atypical, 2 = Non-anginal, 3 = Asymptomatic	0	One-hot encoded
resting_bp	Resting systolic blood pressure	Continuous	94–200 mmHg	0	Z-score standardised
cholesterol	Serum total cholesterol	Continuous	85–564 mg/dL	0	Z-score standardised
fasting_bs	Fasting blood sugar > 120 mg/dL	Binary	0 = No, 1 = Yes	0	Diabetes risk indicator
resting_ecg	Resting electrocardiographic result	Categorical	0 = Normal, 1 = ST-T abnormality, 2 = LVH	0	One-hot encoded
max_heart_rate	Maximum heart rate during exercise	Continuous	60–202 bpm	0	Z-score standardised; strongest predictor
exercise_angina	Exercise-induced angina	Binary	0 = No, 1 = Yes	0	Symptomatic indicator
st_depression	ST segment depression at peak exercise	Continuous	0.0–6.2 mm	0	Z-score standardised; ischemia indicator
st_slope	ST segment slope at	Categorical	0 = Up, 1 = Flat, 2 =	0	One-hot encoded

Variable	Definition	Type	Range / Categories	Missing	Notes
	peak exercise	al	Down		
target (CHD)	Coronary heart disease present	Binary	0 = Absent, 1 = Present	0	508 positive (55.3%), 410 negative (44.7%)

Appendix B: Implementation Overview

All analysis was implemented in Python 3.11 using scikit-learn for modelling, pandas and NumPy for data manipulation, scipy.stats for hypothesis testing, and matplotlib and seaborn for visualisation. The pipeline consists of eight stages mirrored in the architecture diagram (Figure 1): data ingestion, deduplication and preprocessing, exploratory analysis, feature engineering (one-hot encoding and z-score standardisation), model development with stratified 5-fold cross-validation, model evaluation on the held-out test partition, statistical hypothesis testing, and clinical interpretation including the fairness audit and threshold tuning. All transformations were fit on the training partition only to avoid information leakage. The complete code base is publicly available at <https://github.com/mvinayabhushan/Capstone>, organised into scripts/ for the eight pipeline stages, data/ for the cleaned dataset, figures/ for the figures referenced in this report, and a README documenting the directory structure and replication steps.

Appendix C: Statistical Test Results

Table C1 reports the full hypothesis-test results referenced in Results for RQ1 and RQ3. Continuous comparisons use Welch's two-sample t-test with Cohen's d as the effect size; categorical comparisons use chi-square tests of independence with Cramér's V.

Table C1

Statistical Test Results: CHD-Positive vs CHD-Negative Group Comparisons

Variable	CHD+ mean (SD)	CHD- mean (SD)	Statistic	p-value	Effect size
Maximum heart rate	139.2 (19.6)	161.6 (20.2)	$t(916) = -12.8$	< 0.001	$d = 0.85$ (large)
ST depression	1.51 (1.22)	0.70 (0.98)	$t(916) = -11.3$	< 0.001	$d = 0.75$ (medium-large)
Resting blood pressure	135.5 (17.9)	127.3 (16.6)	$t(916) = -6.1$	< 0.001	$d = 0.41$ (medium)
Age	55.6 (8.8)	51.4 (9.1)	$t(916) = -4.2$	< 0.001	$d = 0.28$ (small)
Serum cholesterol	251.1 (52.3)	240.6 (51.1)	$t(916) = -2.9$	0.004	$d = 0.19$ (small)
Sex (male)	—	—	$\chi^2(1) = 29.4$	< 0.001	$V = 0.18$
Chest pain type	—	—	$\chi^2(3) = 127.3$	< 0.001	$V = 0.37$ (medium-large)
Exercise-induced angina	—	—	$\chi^2(1) = 52.1$	< 0.001	$V = 0.24$
ST slope	—	—	$\chi^2(2) = 24.3$	< 0.001	$V = 0.16$
Resting ECG	—	—	$\chi^2(2) = 18.7$	0.001	$V = 0.14$